

ON EXCHANGEABILITY AND HAUSDORFF MOMENT PROBLEMS OVER K -DIMENSIONAL SIMPLEXES

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Summary: Two classical problems, the Hausdorff moment problem and de Finetti's representation theorem for exchangeable random variables, are considered. We generalize these problems to a g -tuple of k -dimensional simplexes and an infinite sequence of g -fold partially exchangeable random variables in $\{0, 1, \dots, k\}$, respectively. The equivalence between them is then formalized and proven.

1. Introduction

Feller (1966), on page 229 of volume II, proved de Finetti's representation theorem for 2-valued exchangeable sequences via the solution to the Hausdorff moment problem over the unit interval. He has also suggested that this method can be extended to a sequence that assumes a finite number of values. These serve as a first motivation for the current work.

Various connections between these two theories may also be found in the literature. For example, Dale (1983) gave a probabilistic proof of Hausdorff's theorem for double sequences using de Finetti's result for partially exchangeable events and Peng, Dang and Wang (2010) showed that partial exchangeability can be characterized by rectangular complete monotonicity. These relations clearly suggest some kind of equivalence between the two theories and Gupta (1999a) established this link between finite Hausdorff moment problems and finite symmetric probabilities. Further, Gupta (1999b) has established the equivalence between the Hausdorff moment problem over a standard k -dimensional simplex and an infinite sequence of exchangeable random variables taking values in a discrete finite domain. However, to the best of our knowledge, equivalence over the infinite case for partially exchangeable sequences has not been formally stated.

The main result of this paper is to prove that solving the Hausdorff moment problem over a g -tuple of k -dimensional simplexes is equivalent to the representation theorem for an infinite sequence

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of g -fold partially exchangeable random variables that take on values in $\{0, 1, \dots, k\}$. We hope that this result serves as a starting point for further generalizations to more complex domains and establishments of additional cross-theoretical results.

The remainder of the paper is structured as follows. In Section 2, we solve the Hausdorff moment problem over a g -tuple of k -dimensional simplexes by generalizing the method of Dale (1987). By following Aldous (1985) and Bernardo and Smith (1994), we give de Finetti's representation theorem for an infinite sequence of g -fold partially exchangeable random variables, that take on values in $\{0, 1, \dots, k\}$, in Section 3. Section 4 provides the main theorem that establishes the equivalence between the above results and Section 5 gives some discussion on further extension to general bounded domains. Conclusion and discussion for further work are provided in Section 6.

2. Hausdorff Moment Problems

Suppose $\{\mu_n : n = 0, 1, \dots\}$ is a sequence of real numbers. The basic Hausdorff moment problem (HMP) is concerned with the existence of a distribution function F such that

$$\mu_n = \int_0^1 x^n dF(x), \quad n = 0, 1, \dots,$$

that is, $\{\mu_n\}$ is the sequence of moments of some random variable X with distribution function F . Hausdorff (1923) showed that a distribution function F exists, and is unique, with the above property if, and only if, the sequence $\{\mu_n\}$ is *completely monotonic*, that is

$$\mu_n \geq 0, \quad \mu_0 = 1 \quad \text{and} \quad \Delta^r \mu_n \geq 0, \quad n, r = 0, 1, \dots$$

where $\Delta \mu_n = \mu_n - \mu_{n+1}$. The conditions above are called the *Hausdorff conditions*. In fact, Hausdorff showed that the conditions can be extended to moment problems over any finite interval on the real line, i.e., for intervals $[a, b]$ such that $-\infty < a < b < \infty$.

Various extensions and modifications of this problem has been published since the above result. One paper of particular interest to us is by Dale (1987), where he extended the problem to the standard triangle $\{(x, y) : x \geq 0, y \geq 0, x + y \leq 1\}$ and solved it by using Bernstein polynomials. We will generalize this method, with the addition of Helly-Bray theorem, to solve the HMP over a g -tuple of k -dimensional simplexes. Let us define, for $i = 1, 2, \dots$,

$$\mathbf{x}_i = (x_{i1}, \dots, x_{ik}), \quad \mathbf{n}_i = (n_{i1}, \dots, n_{ik}), \quad \mathbf{x}_i^{\mathbf{n}_i} = x_{i1}^{n_{i1}} \cdots x_{ik}^{n_{ik}}.$$

and

$$\Omega = \{(\mathbf{x}_1, \dots, \mathbf{x}_g) : x_{ij} \geq 0 \text{ and } \sum_{j=1}^k x_{ij} \leq 1\},$$

which represents a g -tuple of k -dimensional simplexes. Also define $\mathbb{N}_0 = \{0, 1, \dots\}$, $\mathbb{N} = \{1, 2, \dots\}$ and $[g] = \{1, \dots, g\}$ for any positive integer g . We have the following theorem:

Theorem 1 For any given sequence $\{\omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} : \mathbf{n}_i \in [\mathbb{N}_0]^k\}$ of real numbers, there exists a distribution function $F(\mathbf{x}_1, \dots, \mathbf{x}_g)$ on Ω such that

$$\omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} = \int_{\Omega} \mathbf{x}_1^{\mathbf{n}_1} \cdots \mathbf{x}_g^{\mathbf{n}_g} dF(\mathbf{x}_1, \dots, \mathbf{x}_g), \quad n_{ij} = 0, 1, \dots,$$

if, and only if,

$$\omega_{\mathbf{0}, \dots, \mathbf{0}} = 1 \text{ and } \delta_1^{r_1} \cdots \delta_g^{r_g} \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} \geq 0 \text{ for all } \mathbf{n}_i \in [\mathbb{N}_0]^k, r_i = 0, 1, \dots,$$

with δ_i defined by

$$\delta_i \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} = \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} - \sum_{j=1}^k \omega_{\mathbf{n}_1, \dots, \mathbf{n}_i + \mathbf{1}_j, \dots, \mathbf{n}_g},$$

where $\mathbf{1}_j$ are k -dimensional vectors with 1 at the j -th position, 0 everywhere else, and $\delta_i^{r_i}$ are simply the r_i -th power of δ_i .

Proof. (Sketches only)

($\Rightarrow$) Simple.

($\Leftarrow$) First, define the Bernstein polynomial $B_{m_1, \dots, m_g}^f(\mathbf{x}_1, \dots, \mathbf{x}_g)$ over g given k -dimensional simplexes by

$$\sum_{\Gamma} f\left(\frac{\mathbf{u}_1}{m_1}, \dots, \frac{\mathbf{u}_g}{m_g}\right) \prod_{i=1}^g \binom{m_i}{u_{i1} \cdots u_{ik}} x_{i1}^{u_{i1}} \cdots x_{ik}^{u_{ik}} (1 - x_{i1} - \cdots - x_{ik})^{m_i - u_{i1} - \cdots - u_{ik}}$$

where $\Gamma = \{(\mathbf{u}_1, \dots, \mathbf{u}_g) : u_{ij} \geq 0, \sum_{j=1}^k u_{ij} \leq m_i\}$ and f is any function defined and bounded on Ω . Then we can show (similar to Dale, 1987) that, as $m_1, \dots, m_g \rightarrow \infty$,

$$\mathbb{E}(B_{m_1, \dots, m_g}^f) \rightarrow \mathbb{E}(f). \quad (1)$$

Now, let us consider

$$p_{\mathbf{n}_1, \dots, \mathbf{n}_g}^{m_1, \dots, m_g} = \left[\prod_{i=1}^g \binom{m_i}{n_{i1} \cdots n_{ik}} \delta_i^{m_i - n_{i1} - \cdots - n_{ik}} \right] \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g}$$

and hence

$$\sum_{\Gamma} \prod_{i=1}^g \prod_{j=1}^k \binom{u_{ij}}{n_{ij}} p_{\mathbf{u}_1, \dots, \mathbf{u}_g}^{m_1, \dots, m_g} = \prod_{i=1}^g \binom{m_i}{n_{i1} \cdots n_{ik}} \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g}.$$

By choosing $n_{ij} = 0$ for all $i \in [g]$ and all $j \in [k]$, we have

$$\sum_{\Gamma} p_{\mathbf{u}_1, \dots, \mathbf{u}_g}^{m_1, \dots, m_g} = 1.$$

Thus we may define random vectors $\mathbf{X}_{1, m_1}, \dots, \mathbf{X}_{g, m_g}$, with $p_{\mathbf{n}_1, \dots, \mathbf{n}_g}^{m_1, \dots, m_g}$ as the joint atomic distribution, i.e.,

$$\mathbb{P}\left(\mathbf{X}_{1, m_1} = \frac{\mathbf{n}_1}{m_1}, \dots, \mathbf{X}_{g, m_g} = \frac{\mathbf{n}_g}{m_g}\right) = p_{\mathbf{n}_1, \dots, \mathbf{n}_g}^{m_1, \dots, m_g}.$$

By applying result (1), Helly's theorem (see Feller, 1966) and the Helly-Bray theorem (see Tucker, 1967) we obtain

$$\omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} = \mathbb{E}(\mathbf{X}_1^{\mathbf{n}_1} \cdots \mathbf{X}_g^{\mathbf{n}_g}),$$

where $\mathbf{X}_1, \dots, \mathbf{X}_g$ are random vectors distributed according to the limit distribution $p_{\mathbf{n}_1, \dots, \mathbf{n}_g}$ over Ω , when $m_1, \dots, m_g$ tend to infinity. This completes the proof of Theorem 1. $\blacksquare$

It is easy to see that, for particular choices of g and k , the above theorem reduces to Hausdorff moment problems over the unit line, the unit square and the standard triangle. In fact, further refinements of the above result are possible, which is discussed in Section 5.

3. Exchangeability and de Finetti's Representation Theorems

In 1930, exchangeable random variables were introduced by de Finetti in the context of personalistic probability specification (see de Finetti, 1930; 1974). We may state this property as follows:

A sequence of random variables $X_1, \dots, X_n$ is said to be exchangeable if their joint distribution function F satisfies

$$F(x_1, \dots, x_n) = F(x_{\pi(1)}, \dots, x_{\pi(n)}),$$

for all permutations π defined on the set $\{1, 2, \dots, n\}$.

In terms of the corresponding density or mass function, the condition reduces to

$$p(x_1, \dots, x_n) = p(x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(n)}),$$

where $p(\cdot)$ is the joint density or mass function. Moreover, an infinite sequence of random variables $\{X_n\}$ is said to be exchangeable if all of its finite subsequences are exchangeable.

De Finetti then generalized the concept of exchangeability to “partial exchangeability”, which considers a sequence of several types being exchangeable within each type (see de Finetti, 1938), i.e., a g -fold sequence of random variables, $\{X_{ij} : i \in [g], j \in [n_i]\}$, is said to be *g -fold partially exchangeable* if the joint distribution of the $n_1, \dots, n_g$ random variables of types $1, \dots, g$ respectively, depends only on $n_1, \dots, n_g$ and not on the order of the random variables within each type. Therefore, an infinite g -fold sequence of random variables is said to be g -fold partially exchangeable if all its finite g -fold subsequences of random variables are partially exchangeable.

In the papers, de Finetti also gave his famous representation theorem for the 2-valued case, which showed that the joint distribution of any 2-valued exchangeable sequence can be represented as a mixture of IID sequence of random variables. In the theorem below, this result is extended to infinite sequences of partially exchangeable random variables that take on finitely many values.

Theorem 2 An infinite sequence of random variables $\{X_{ij} : i \in [g], j \in \mathbb{N}\}$, taking values in $\{0, 1, \dots, k\}$, is g -fold partially exchangeable if, and only if, there exists a distribution function F such that the probability of obtaining n_{i1} 1's, n_{i2} 2's, ... and n_{ik} k 's out of m_i variables in the i -th within-type sequence, for all $i \in [g]$, is given by

$$\int_{\Omega} \prod_{i=1}^g \binom{m_i}{n_{i1} \dots n_{ik}} x_{i1}^{n_{i1}} \dots x_{ik}^{n_{ik}} (1 - \sum_j x_{ij})^{m_i - \sum_j n_{ij}} dF(\mathbf{x}_1, \dots, \mathbf{x}_g),$$

where x_{ij} is the probability of obtaining j in the i -th within-type sequence, denoted by $\{X_{ij} : j \in \mathbb{N}\}$, and Ω is defined as before.

Proof. The proof is a straightforward generalization of the 2-valued result given in Bernardo and Smith (1994), by considering g urns that contain $(k+1)$ types of items, out of the total of m_i items in each urn. Then obtaining a product of multinomial expressions over the g urns. See also Aldous (1985). ■

It is also easy to note that the above theorem is independent of the choice of the set $\{0, 1, \dots, k\}$, meaning we could have chosen any arbitrary set that contained $(k+1)$ points.

4. Equivalence over a g -tuple of Simplexes

We now show that solving the HMP over a g -tuple of k -dimensional simplexes is equivalent to a g -fold infinite sequence of random variables, that take on any $(k+1)$ values, being partially exchangeable (i.e., the representation theorem for infinite sequences of partially exchangeable random variables). The proof given below is a further generalization of Gupta (1999b), with the added notion of partial exchangeability, but the route we took is also easier in the sense that we do not introduce infinite product probabilities, nor probability on a class of probabilities. Rather, we explore the simple probabilistic implications of the given distribution.

Theorem 3 An infinite sequence of random variables $\{X_{ij} : i \in [g], j \in \mathbb{N}\}$, taking values in $\{0, 1, \dots, k\}$, is g -fold partially exchangeable if, and only if, the corresponding HMP over a g -tuple of k -dimensional simplexes has a solution, i.e., there exists a distribution function F such that, for Ω defined as before and all $n_{ij} \in \mathbb{N}_0$, we have

$$\omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} = \int_{\Omega} \mathbf{x}_1^{\mathbf{n}_1} \cdots \mathbf{x}_g^{\mathbf{n}_g} dF(\mathbf{x}_1, \dots, \mathbf{x}_g),$$

which is equal to the probability of having the first n_{i1} of $\{X_{ij} : j \in \mathbb{N}\}$ being 1's, the next n_{i2} of $\{X_{ij} : j \in \mathbb{N}\}$ being 2's, ... etc., and the last n_{ik} of $\{X_{ij} : j \in \mathbb{N}\}$ being k 's, for all $i \in [g]$.

Proof. ($\Rightarrow$) Let $\{X_{ij} : i \in [g], j \in \mathbb{N}\}$ be a g -fold partially exchangeable sequence of random variables taking values in $\{0, 1, \dots, k\}$ and define $p_{\mathbf{n}_1, \dots, \mathbf{n}_g, \mathbf{m}}$ as the probability that, for each i , the first n_{i1} X_{ij} 's equal 1, followed by n_{i2} 2's, ... etc., and the last $m_i - n_{i1} - \dots - n_{ik}$ X_{ij} 's equal to 0, for all $n_{i1} + \dots + n_{ik} \leq m_i$. Also define

$$\omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} = p_{\mathbf{n}_1, \dots, \mathbf{n}_g, \sum_i \mathbf{n}_i \mathbf{A}_i}$$

for all $n_{ij} = 0, 1, \dots$, where $\mathbf{A}_i$ are the matrices with 1's in the i -th column and 0 everywhere else, and put $\omega_{\mathbf{0}, \dots, \mathbf{0}} = 1$. Then we get

$$\begin{aligned} p_{\mathbf{n}_1, \dots, \mathbf{n}_g, \sum_i \mathbf{n}_i \mathbf{A}_i + \mathbf{1}_1} &= p_{\mathbf{n}_1, \dots, \mathbf{n}_g, \sum_i \mathbf{n}_i \mathbf{A}_i} - p_{\mathbf{n}_1 + \mathbf{1}_1, \dots, \mathbf{n}_g, \sum_i \mathbf{n}_i \mathbf{A}_i + \mathbf{1}_1} \\ &\quad - p_{\mathbf{n}_1 + \mathbf{1}_2, \mathbf{n}_2, \dots, \mathbf{n}_g, \sum_i \mathbf{n}_i \mathbf{A}_i + \mathbf{1}_1} \\ &\quad - \dots - p_{\mathbf{n}_1 + \mathbf{1}_k, \dots, \mathbf{n}_g, \sum_i \mathbf{n}_i \mathbf{A}_i + \mathbf{1}_1} \\ &= \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} - \omega_{\mathbf{n}_1 + \mathbf{1}_1, \dots, \mathbf{n}_g} - \dots - \omega_{\mathbf{n}_1 + \mathbf{1}_k, \dots, \mathbf{n}_g} \\ &= \delta_1 \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g}, \end{aligned}$$

and similarly we can get $p_{\mathbf{n}_1, \dots, \mathbf{n}_g, \sum_i \mathbf{n}_i \mathbf{A}_i + \mathbf{1}_j} = \delta_j \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g}$, for δ_j and $\mathbf{1}_j$ defined as in Theorem 1. Then, by induction, we get,

$$p_{\mathbf{n}_1, \dots, \mathbf{n}_g, \sum_i \mathbf{n}_i \mathbf{A}_i + \mathbf{r}} = \delta_1^{r_1} \delta_2^{r_2} \cdots \delta_g^{r_g} \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g},$$

which is obviously greater than or equal to zero for any $\mathbf{r} = (r_1, \dots, r_g) \in [\mathbb{N}_0]^g$, i.e. $\{\omega_{\mathbf{n}_1, \dots, \mathbf{n}_g}\}$ is a moment sequence (by Theorem 1) defined by a distribution F over Ω , as defined in the theorem statement, such that

$$\omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} = \int_{\Omega} \mathbf{x}_1^{\mathbf{n}_1} \cdots \mathbf{x}_g^{\mathbf{n}_g} dF(\mathbf{x}_1, \dots, \mathbf{x}_g).$$

($\Leftarrow$) Suppose that $\{\omega_{\mathbf{n}_1, \dots, \mathbf{n}_g}\}$ is a moment sequence over a g -tuple of k -dimensional simplexes, with distribution function F , then we have

$$\left[\prod_{i=1}^g \delta_i^{m_i - n_{i1} - \dots - n_{ik}} \right] \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} \geq 0,$$

for all $\mathbf{m} \in [\mathbb{N}_0]^g$ such that $n_{i1} + \dots + n_{ik} \leq m_i$, and we may show

$$\begin{aligned} & \sum_{\mathbf{n}_1, \dots, \mathbf{n}_g \in Q} \left[\prod_{i=1}^g \delta_i^{m_i - n_{i1} - \dots - n_{ik}} \binom{m_i}{n_{i1} \dots n_{ik}} \right] \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} \\ &= \sum \int_{\Omega} \left[\prod_{i=1}^g \binom{m_i}{n_{i1} \dots n_{ik}} x_{i1}^{n_{i1}} \dots x_{ik}^{n_{ik}} (1 - \sum_j x_{ij})^{m_i - \sum_j n_{ij}} \right] dF \\ &= 1 \end{aligned}$$

for $Q = \{(\mathbf{n}_1, \dots, \mathbf{n}_g) : n_{ij} \geq 0, \sum_j n_{ij} \leq m_i\}$. We may then define a sequence $\{X_{ij} : i \in [g], j \in \mathbb{N}\}$, taking values in $\{0, 1, \dots, k\}$, such that

$$\begin{aligned} & \mathbb{P} \left(\begin{array}{l} \text{for each } i = 1, \dots, g, \text{ a particular choice of } n_{ij} \text{ of } X_{ij} \text{'s} \\ \text{equal to } j, \text{ for all } j, \text{ from the total of } m_i \text{ } X_{ij} \text{'s} \end{array} \right) \\ &= \left[\prod_{i=1}^g \delta_i^{m_i - n_{i1} - \dots - n_{ik}} \right] \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g} \end{aligned}$$

and thus

$$\begin{aligned} & \mathbb{P} \left(\begin{array}{l} \text{for each } i = 1, \dots, g, \text{ any choice of } n_{ij} \text{ of } X_{ij} \text{'s} \\ \text{equal to } j, \text{ for all } j, \text{ from the total of } m_i \text{ } X_{ij} \text{'s} \end{array} \right) \\ &= \left[\prod_{i=1}^g \delta_i^{m_i - n_{i1} - \dots - n_{ik}} \binom{m_i}{n_{i1} \dots n_{ik}} \right] \omega_{\mathbf{n}_1, \dots, \mathbf{n}_g}. \end{aligned}$$

Then, clearly, $\{X_{ij} : i \in [g], j \in \mathbb{N}\}$ is a g -fold partially exchangeable sequence. This completes the proof for Theorem 3. $\blacksquare$

The above result clearly covers all particular cases of infinite exchangeable, and partially exchangeable, sequences of random variables that take on finitely many values.

5. Extensions to General Bounded Domains

Stockbridge (2003) provided the solutions to HMP over polytopes

$$B = \{(\theta_1, \dots, \theta_k) : a_{h1}\theta_1 + \dots + a_{hk}\theta_k \leq a_{h0}, h = 1, \dots, g\}$$

and more general bounded regions defined by

$$C = \{(\theta_1, \dots, \theta_k) : a_{h1}\theta_1^{b_{h1}} + \dots + a_{hk}\theta_k^{b_{hk}} \leq a_{h0}, h = 1, \dots, g\}.$$

We can deal with these cases through defining the increments and the p terms differently, then we may obtain the solutions in similar ways as in Theorem 3. For example, in the basic case of the HMP over $[0, b]$, we define the increments by

$$\Delta\mu_n = b\mu_n - \mu_{n+1}$$

which will eventually lead to

$$\sum_{k=0}^n \binom{n}{k} \left(\frac{1}{b}\right)^n \Delta^{n-k} \mu_k = 1.$$

So we can define an exchangeable sequence $X_1, X_2, \dots$ in $\{0, 1\}$ by

$$\mathbb{P}(k \text{ out of } n \text{ } X_i\text{'s are equal to 1}) = \binom{n}{k} \left(\frac{1}{b}\right)^n \Delta^{n-k} \mu_k.$$

Conversely, define

$$p_{k,n} = b^n \mathbb{P}(X_1 = 1, \dots, X_k = 1, X_{k+1} = 0, \dots, X_n = 0)$$

and

$$\mu_n = p_{n,n} = \mathbb{P}(X_1 = 1, \dots, X_n = 1)$$

then we get

$$p_{n-1,n} = bp_{n-1,n-1} - p_{n,n} = \Delta\mu_{n-1}$$

and hence

$$p_{n-2,n} = bp_{n-2,n-1} - p_{n-1,n} = \Delta^2 \mu_{n-2}.$$

Therefore, by induction, we have for all $k < n$

$$p_{k,n} = bp_{k,n-1} - p_{k+1,n} = \Delta^{n-k} \mu_k.$$

So, this means that we obtain the same result for HMP over $[0, b]$ as for HMP over $[0, 1]$. A similar approach can be applied to HMP over polytopes and more general bounded regions defined above.

6. Discussion and Further Research

In this paper, we have generalized the Hausdorff moment problem to a g -tuple of k -dimensional simplexes and stated de Finetti's representation theorem for a g -fold infinite sequence of partially exchangeable random variables taking values in $\{0, 1, \dots, k\}$. And then, we formally established the equivalence between the two. Furthermore, we have shown that this equivalence can be easily extended to general bounded domains in the sense of Stockbridge (2003).

Finite Hausdorff moment problems refer to cases where only a finite number of moments are available. Likewise, one may consider a sequence of only finitely many exchangeable random variables taking values in a finite discrete domain. Equivalence under these cases were dealt with by Gupta (1999a). In particular, he established the links between finite Hausdorff moment problems and finite symmetric probabilities. However, a possible interest for future research may be

to look at translating the various approximation procedures for finite moment problems (such as Talenti, 1987; Inglese, 1995) to finite exchangeable sequences, and vice-versa.

Extensions to exchangeable sequences that take on values in a continuous domain are much more complicated. This is due to the fact that the corresponding representation theorem result in an integral defined over the set of all distribution functions in the domain of interest and a probability measure over the set of all probability measures in the same domain (see, for example, Aldous, 1985). Insights into the extensions for such cases would be of interest for further exploration.

It is also somewhat obvious that we have been dealing with HMP over bounded regions because we are, in essence, looking for distribution functions and probability measures which are clearly bounded. In theory, one may consider general measures that need not be bounded and this could relate to more general types of moment problems over unbounded domains, for example, the Stieltjes moment problem over $[0, \infty)$ and the Hamburger moment problem over $(-\infty, \infty)$. It would be interesting to see how the two concepts of exchangeability and moment problems may connect or disconnect under these cases. These are left, perhaps, for future research.

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